

Mount Kenya University

Unlocking Infinite Possibilities

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COURSE: COMPUTER SCIENCE

UNIT CODE: BIT3208

UNIT NAME: ADVANCED WEB DESIGN AND DEVELOPMENT

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SEMESTER: 3.2

PROJECT TITLE: AN E-COMMERCE PLATFORM

SELECTED TECHNOLOGIES: PHP + MySQL

GITHUB LINK: <https://github.com/Coutts007/Advancedweb>

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WEEK 1: LOCAL ENVIRONMENT SETUP

Title: Installation and Testing of Local Development Environment

Fig 1: Installation of XAMPP

-The figure below shows XAMPP installed in my device.

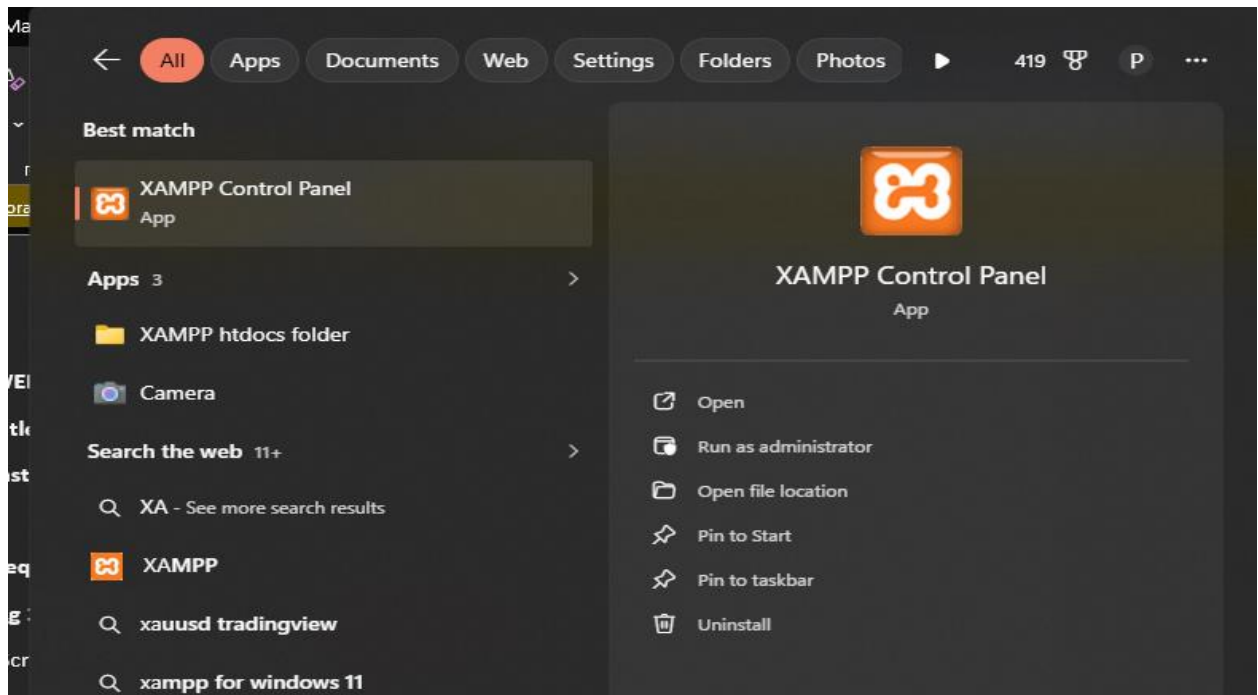


Fig 2: Apache and MySQL Running

-It shows the XAMPP control panel with the Apache and MySQL running successfully.

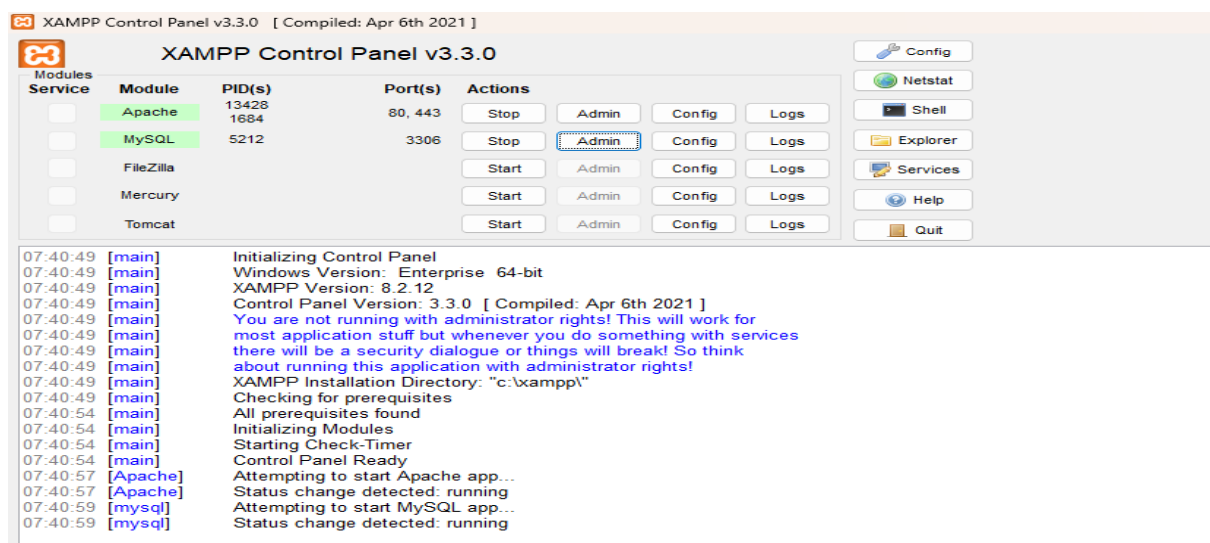


Fig 3: Localhost Test Page

-Below figure shows a successfully running localhost server.

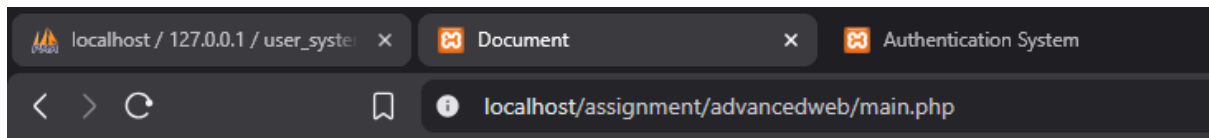
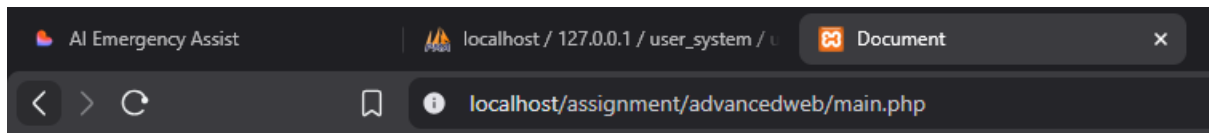


Fig 4: Hello World Test

-The figure below shows first php file executed successfully in the local sever displaying “Hello, world!”

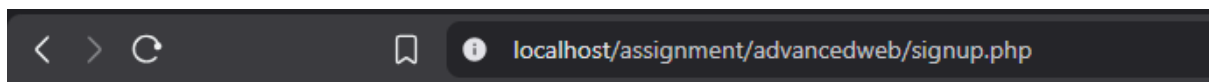


Hello, World!

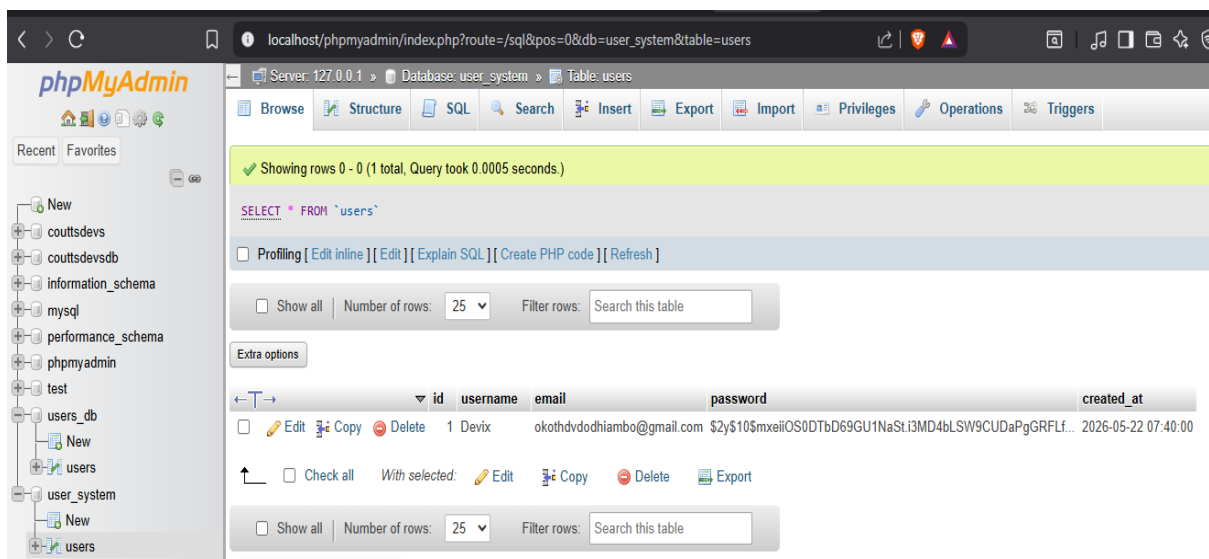
Welcome to My Website

Fig 5: Database Connection Test

-Shows successful connection between PHP and MySQL using mysqli_connect(). This confirmed that Apache and MySQL services were properly configured in XAMPP.



Registration Successful!



REFLECTION.

I learnt how to configure a PHP localhost server that ran successfully and also to create a successful database connection for my web page using PHP.

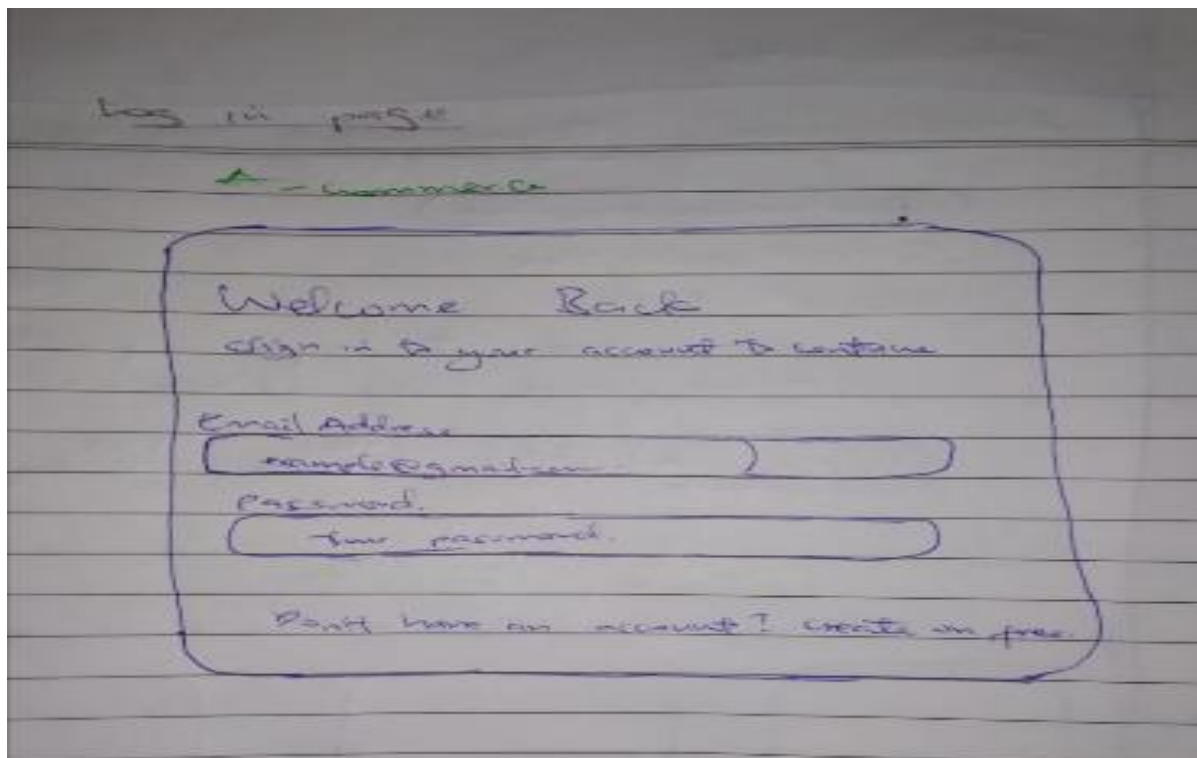
The technologies I used are XAMPP and MySQL for database connection for my web page.

WEEK 2: WIREFRAMES AND GUI DESIGN

Title: User Interface Planning and System Design

Fig 1: Hand-drawn Wireframe

-The below figures show the website login and homepage designs respectively.



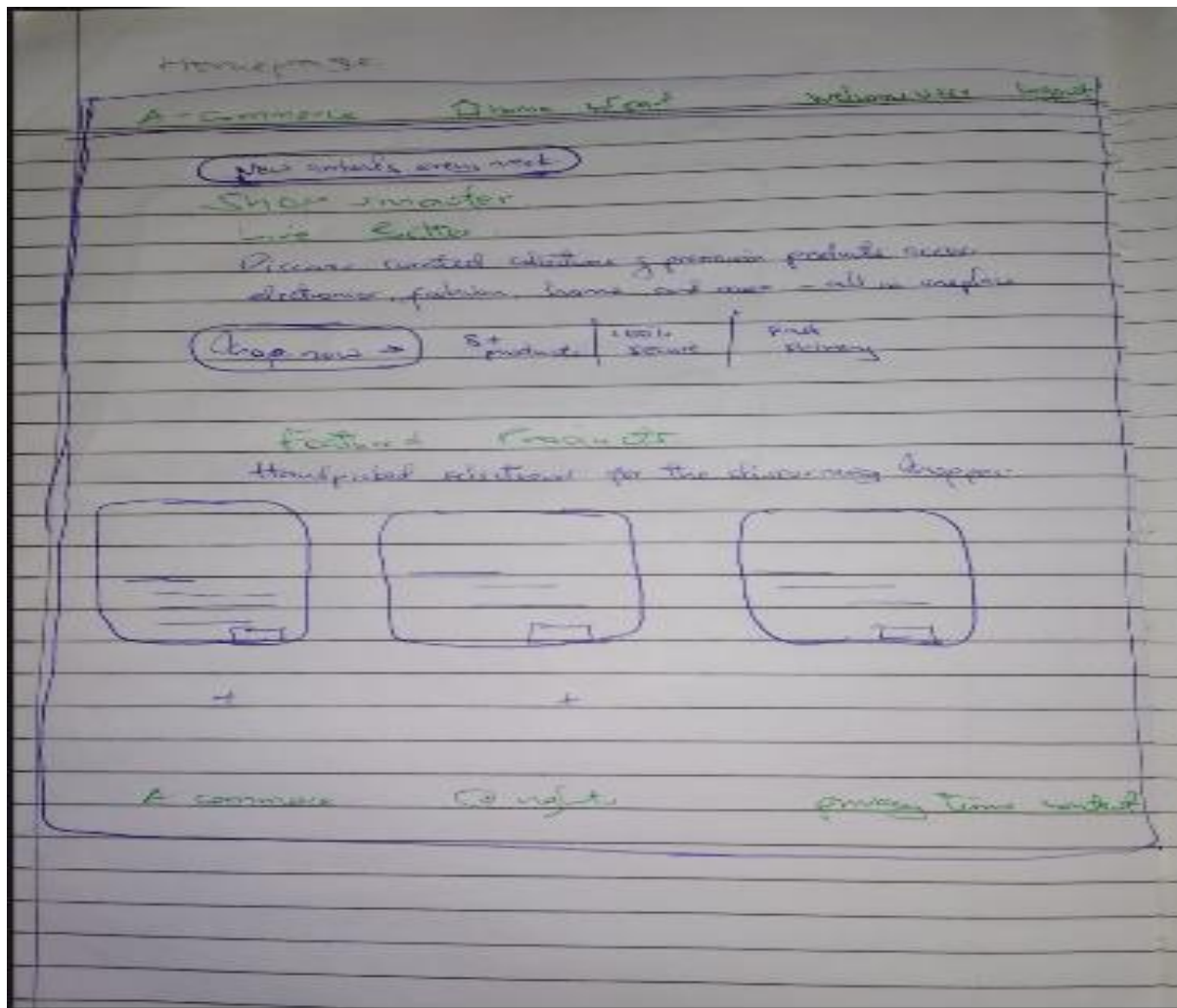


Fig 2: Dashboard Layout Design

-Fig 2 below is a Homepage design implementation for the ecommerce website from the sketch above.

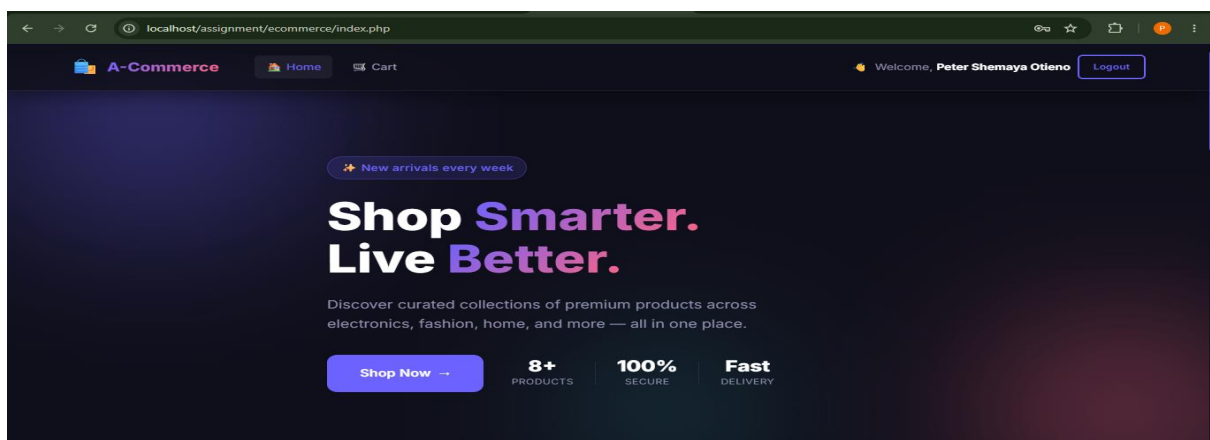


Fig 3: Mobile View Mock-up

-The below figure shows a mobile view of the implemented homepage.

-The navigation modifies into a single menu point to adapt to screen size change.

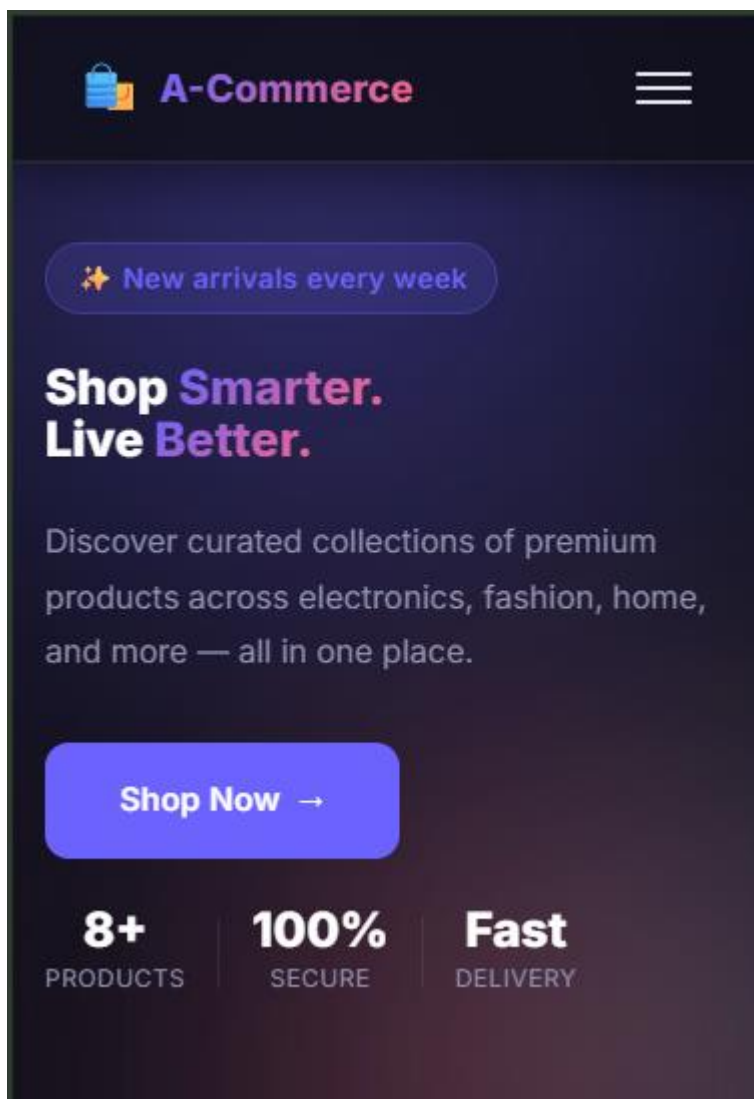
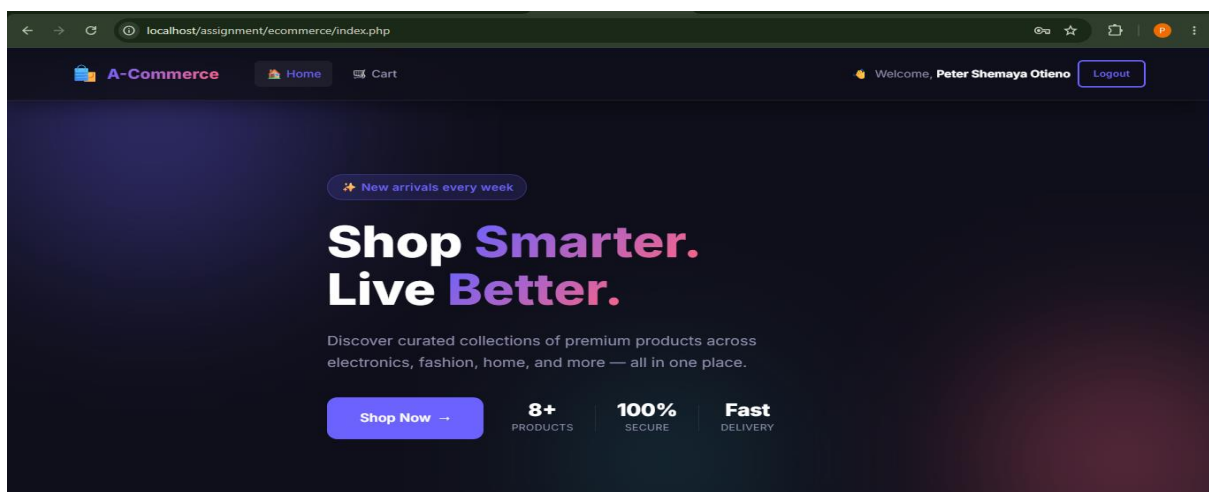


Fig 4: Color Theme and Navigation Structure



Colour Theme

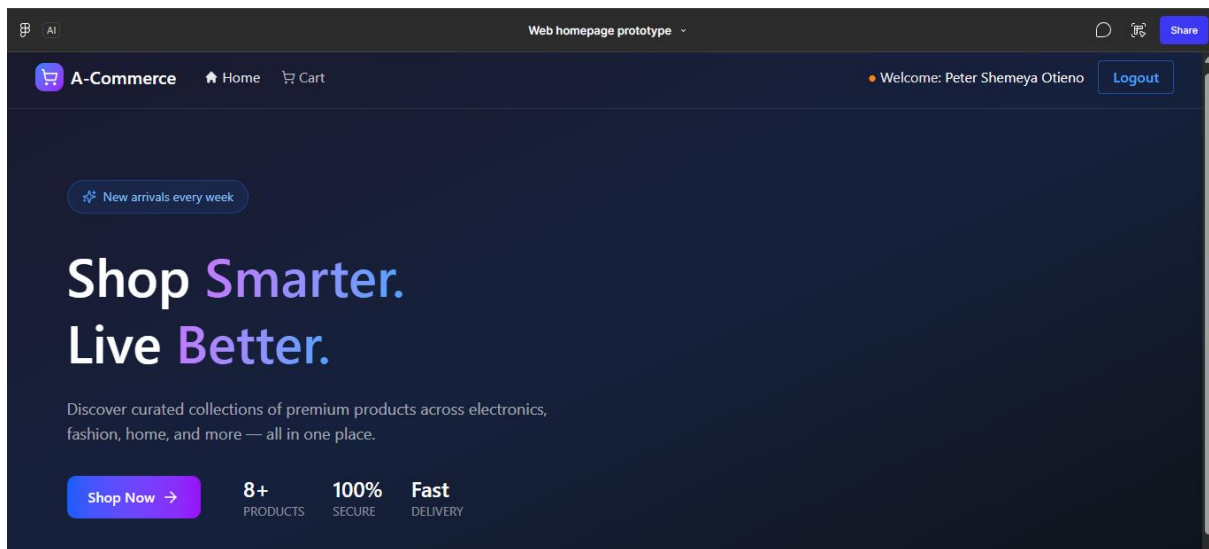
- Vibe: Dark mode with a sleek, modern gradient.
- Background: Deep navy-blue header blending into a dark purple hero section.
- Text: Bright white for headlines and key stats; muted grey for the descriptive text.

Navigation

- Layout: A clean, minimal top header.
- Menu: keeping the mobile layout tidy by hiding the main links until clicked.

Fig 5: GUI Prototype in Figma

-Below is a the Figma homepage prototype for the website.



The homepage layout was designed to reduce user clicks and improve accessibility. The navigation menu was positioned on the left for easier scalability as more modules are added.

WEEK 3: JAVASCRIPT AND PHP BASICS

Title: Frontend Interaction and Backend Foundations

Fig 1: JavaScript Form Validation

-The fig 1 below shows JavaScript form validation code. It ensures the form fields required are all filled before submission.


```

186
187 // — Client-side form validation —
188 const form      = document.getElementById('signup-form');
189 const jsError    = document.getElementById('js-error');
190 const jsErrorText = document.getElementById('js-error-text');
191 const signupBtn  = document.getElementById('signup-btn');
192
193 function showError(msg) {
194     jsErrorText.textContent = msg;
195     jsError.style.display   = 'flex';
196     jsError.scrollIntoView({ behavior: 'smooth', block: 'nearest' });
197 }
198
199 function hideError() {
200     jsError.style.display = 'none';
201 }
202
203 // Validate email format
204 function isValidEmail(email) {
205     return /^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(email.trim());
206 }
207
208 form.addEventListener('submit', (e) => {
209     hideError();
210
211     const name      = document.getElementById('name').value.trim();
212     const email      = document.getElementById('email').value.trim();
213     const password    = document.getElementById('password').value;
214     const confirm     = document.getElementById('confirm_password').value;

```

Fig 2: Password Strength Checker

The figure below shows code section that implements password strength checking before a user signs up.

```

151 <script>
152 // — Password strength meter —
153 const passwordInput = document.getElementById('password');
154 const strengthFill   = document.getElementById('strength-fill');
155 const strengthLabel  = document.getElementById('strength-label');
156
157 function measureStrength(pw) {
158     let score = 0;
159     if (pw.length >= 8) score++;
160     if (pw.length >= 12) score++;
161     if (/[A-Z]/.test(pw)) score++;
162     if (/[\d-9]/.test(pw)) score++;
163     if (/^[^A-Za-z0-9]/.test(pw)) score++;
164     return score;
165 }
166
167 passwordInput.addEventListener('input', () => {
168     const score = measureStrength(passwordInput.value);
169     const pct   = (score / 5) * 100;
170     const levels = ['', 'Very Weak', 'Weak', 'Fair', 'Strong', 'Very Strong'];
171     const colors = ['', '#e74c3c', '#e67e22', '#f1c40f', '#2ecc71', '#27ae60'];
172     strengthFill.style.width = pct + '%';
173     strengthFill.style.background = colors[score] || '#ddd';
174     strengthLabel.textContent = score > 0 ? levels[score] : '';
175     strengthLabel.style.color = colors[score] || '#aaa';
176 });
177

```

Fig 3: PHP Syntax Practice

```

C: > xampp > htdocs > assignment > Class > learn.php > ...
1  <?php
2  $Electronics = array("TV", "Phone", "Radio", "DVD", "CD-ROM");
3  for ($i = 0; $i < count($Electronics);
4  $i++){
5      echo "$Electronics[$i]";
6  }
7  foreach ($Electronics as $electronic ){
8      echo $electronic;
9  }
10
11 ?>

```

Fig 4: Database Connection Script

-The script below connects the website to the database.

```
19 // — Collect & sanitize inputs —————
20 $name      = trim(htmlspecialchars($_POST['name']      ?? '', ENT_QUOTES, 'UTF-8'));
21 $email     = trim(htmlspecialchars($_POST['email']     ?? '', ENT_QUOTES, 'UTF-8'));
22 $password  = $_POST['password'] ?? '';
23 $confirm_password = $_POST['confirm_password'] ?? '';
24
25 // — Server-side validation —————
26 $errors = [];
27
28 if (empty($name) || mb_strlen($name) < 2 || mb_strlen($name) > 120) {
29     $errors[] = 'Name must be between 2 and 120 characters.';
30 }
31
32 if (!filter_var($email, FILTER_VALIDATE_EMAIL) || mb_strlen($email) > 180) {
33     $errors[] = 'Please provide a valid email address.';
34 }
35
36 if (mb_strlen($password) < 8) {
37     $errors[] = 'Password must be at least 8 characters long.';
38 }
39
40 if ($password !== $confirm_password) {
41     $errors[] = 'Passwords do not match.';
42 }
43
44 if (!empty($errors)) {
45     setFlash('error', implode(' ', $errors));
46     redirect('signup.php');
47 }
```

Fig 5: Dynamic User Input Handling

-The code below enables efficient and dynamic addition of products in the website.

```
30 CREATE TABLE IF NOT EXISTS products (
31     id          INT          NOT NULL AUTO_INCREMENT,
32     title       VARCHAR(200) NOT NULL,
33     description TEXT         NOT NULL,
34     price       DECIMAL(10, 2) NOT NULL,
35     image_url   VARCHAR(500) NOT NULL,
36     stock       INT          NOT NULL DEFAULT 0,
37     created_at  TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP,
38     PRIMARY KEY (id),
39     CONSTRAINT chk_price CHECK (price >= 0),
40     CONSTRAINT chk_stock  CHECK (stock >= 0)
41 ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
42
43 -- =====
44 -- 4. Seed - 6 realistic mock products
45 -- Images sourced from picsum.photos (deterministic, stable IDs)
46 -- =====
47 INSERT INTO products (title, description, price, image_url, stock) VALUES
48
49 ('Sony WH-1000XM5 Headphones',
50  'Industry-leading noise cancellation with 30-hour battery life. Crystal-clear call quality with eight microphones. Ultra-comfortable',
51  349.99,
52  'https://media.s-bol.com/gMLJEBpA8NpZ/w0kkgoX/779x1200.jpg',
53  42),
54
55 ('Apple iPad Pro 11" (M4)',
56  'Supercharged by the Apple M4 chip with an Ultra Retina XDR OLED display. Incredibly thin at just 5.1mm. Supports Apple Pencil Pro',
57  999.00,
58  'https://assets.swappie.com/cdn-cgi/image/width=600,height=600,dpr=2,fit=contain,format=auto/swappie-ipad-pro-3-2021-11-space-gra',
59  18),
```

WEEK 4: SERVER-SIDE COMPONENTS AND BACKEND DEVELOPMENT

FOUNDATIONS

Fig 4.1: Modular Folder Structure.

-The below figure shows a clear structured project folder.

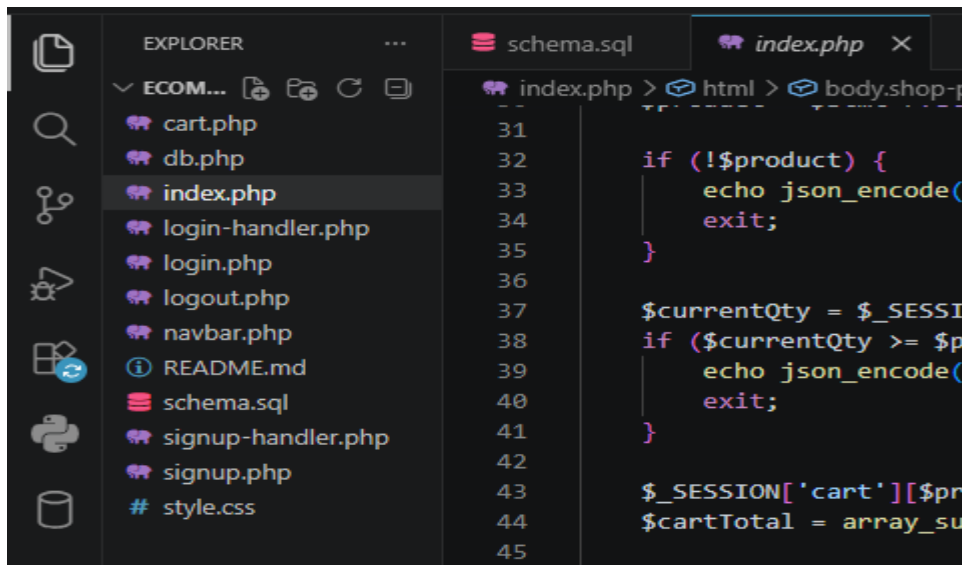
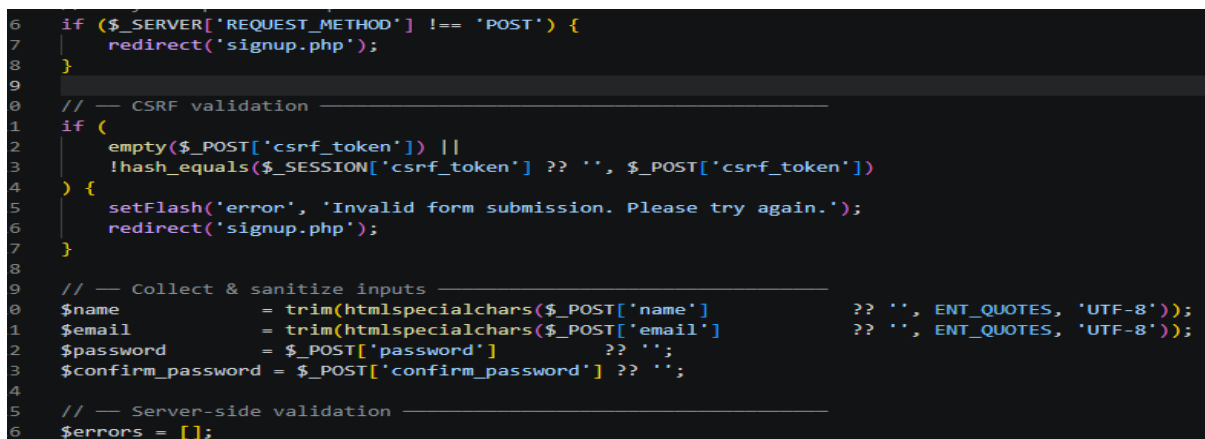


Fig 4.2: PHP Form Processing. Fig 4.3:



WEEK 5: DATABASE COMPONENTS AND CRUD OPERATIONS

Implementing Secure Database Management and Access Control

Fig 5.1: Database Schema Expansion.

 		 id	name	email	password	is_admin	created_at		
☐	 Edit	 Copy	 Delete	1	Admin User	admin@acommerce.local	\$2y\$10\$UD.OwkFVQn8SXRFPwGaBout1zHNehTxtOy4fKF7jL0j..	1	2026-06-04 14:44:54
☐	 Edit	 Copy	 Delete	2	John Customer	customer@acommerce.local	\$2y\$10\$NJKw1ffUGSAcZWR/H3oRcudREnioPMsLZOS.GjSxwly..	0	2026-06-04 14:44:54
☐	 Edit	 Copy	 Delete	3	Peter Shemaya	petershemaya007@gmail.com	\$2y\$12\$1jSwAoB5fproEJoWnzidVe3ZiGQQqMfJAICgXjgVhNf...	0	2026-06-04 14:52:52
☐	 Edit	 Copy	 Delete	4	David Okoth	okothdvdodhiambo@gmail.com	\$2y\$12\$PISuBwQYZLpjD3ig9ZVmiOQ5AICH.2Cq904L3oppKzZ...	0	2026-06-05 11:16:29
☐	 Edit	 Copy	 Delete	5	Annitah Santuri	annitahsanturi@gmail.com	\$2y\$12\$mVXhJPETPY1WHiVpf8YmluWZL/gsOUUnwEilluR7DHZF	0	2026-06-05 11:18:19
	☐ Check all	With selected:	 Edit	 Copy	 Delete	 Export			

Fig 5.2: The CRUD Administrative Panel.

A-Commerce

Home

Cart

Admin

Welcome, **Admin User** [Logout](#)

Recent Orders

ORDER ID	CUSTOMER	ITEMS	TOTAL	DATE
#3	John Customer	1	\$349.99	Jun 02, 2026
#2	John Customer	2	\$239.94	May 30, 2026
#1	John Customer	2	\$484.94	May 28, 2026

Top Products

PRODUCT	PRICE	UNITS SOLD	REVENUE
Sony WH-1000XM5 Headphones	\$349.99	2	\$699.98
Instant Pot Duo 7-in-1 Electric Pressure Cooker	\$89.95	1	\$89.95

Fig 5.3: Role-Based UI Rendering.

-The figure below shows admin and users having different access rights on their dashboards.

A-Commerce

Home

Cart

Admin

Welcome, **Admin User** [Logout](#)

Admin Dashboard

Manage products and monitor sales

A-Commerce

Home

Cart

Welcome, **Peter Shemaya** [Logout](#)[New arrivals every week](#)

Fig 5.4: Secure Server Logic.

```
1  <?php
2  // login-handler.php – Authentication backend processor
3  require_once 'db.php';
4
5  // Only accept POST requests
6  if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
7      redirect('login.php');
8  }
9
10 // — CSRF validation —————
11 if (
12     empty($_POST['csrf_token']) ||
13     !hash_equals($_SESSION['csrf_token'] ?? '', $_POST['csrf_token'])
14 ) {
15     setFlash('error', 'Invalid form submission. Please try again.');
```

-So far up to week 5 is an implementation of a functional e commerce platform with a server logic. It consists of admin dashboard and client dashboard which have different rights to the system.